THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A body moving in a circle of radius r completes one revolution in time T. Its speed is: 1

(a) pi r / T **(b)** 2 pi r / T **(c)** 2 r / T **(d)** r / T

2. A boy walks 3 m east and then 4 m north. The magnitude of his displacement is: 1

(a) 1 m **(b)** 12 m **(c)** 7 m **(d)** 5 m

3. The average speed of a car which travels 60 km in 1 hour and then 40 km in the next 1 hour is: 1

(a) 60 km/h **(b)** 50 km/h **(c)** 100 km/h **(d)** 40 km/h

4. A car's speedometer shows 60 km/h. This reading is its: 1

(a) average velocity **(b)** acceleration
(c) instantaneous speed **(d)** displacement

5. A body moves with uniform velocity if it covers: 1

(a) equal displacements in equal intervals of time in the same direction
(b) unequal distances
(c) equal distances in equal times in a circle
(d) zero distance

6. Which equation is correct for uniformly accelerated motion? 1

(a) v = u - at only **(b)** v = at squared **(c)** v = u + at **(d)** s = vt always

7. Negative acceleration is called: 1

(a) inertia **(b)** velocity
(c) speed **(d)** retardation (deceleration)

8. The slope of a distance-time graph gives: 1

(a) force **(b)** acceleration **(c)** displacement **(d)** speed

9. Which equation is correct for uniformly accelerated motion? 1

(a) s = ut squared **(b)** s = ut + (1/2)at squared
(c) s = ut - at **(d)** s = u + at

10. An object is said to be in motion when its position: 1

(a) changes with time relative to a reference point
(b) is zero
(c) is at origin
(d) does not change

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is meant by a uniform and non-uniform motion? Give one example each. 2

12. Convert 72 km/h into m/s. 2

13. A car increases its speed from 10 m/s to 20 m/s in 5 s. Find its acceleration. 2

14. What is average speed? A man runs 100 m in 20 s. Find his speed. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A ball is dropped from a height and takes 3 s to reach the ground. Find the height and the velocity just before hitting the ground (g = 10 m/s squared). 3

16. A body travels 20 m in the first 2 s and 30 m in the next 3 s. Find the average speed. 3

17. Explain what a position-time graph that is a curve bending upward tells us about the motion. 3

OR

Draw in words the distance-time graph of a body moving with uniform speed and show how to find the speed. 3

18. Derive v = u + at from the definition of acceleration. 3

OR

Name three types of motion with examples of each. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the distance-time and velocity-time graphs for 5

(a) rest
(b) uniform velocity
(c) uniform acceleration
(d) uniform retardation.

OR

Explain the importance of the frame of reference with examples: passengers in a moving train and a person on the platform. 5

20. A cyclist rides with a speed of 18 km/h, accelerates uniformly at 0.5 m/s squared for 10 s and then applies brakes to stop in 20 s. Find the final speed after acceleration, the retardation and the total distance. 5

OR

A motorist travels 120 km at 60 km/h and returns at 40 km/h. Find the average speed for the whole trip and explain why it is not 50 km/h. 5

END OF QUESTION PAPER Total: 40 Marks